Diameter in graphs

Very simple but useful heuristics: 2-SWEEP

Invariant At any time, we maintain lower bound L and upper bound U, such that $L \leq D \leq U$ for the diameter D

When $L = U \Rightarrow$ we have found D

IDEA Using BFS, try to refine both L and U:
- try to increase L $(L=0)$
- try to decrease U $(U=n-1)$

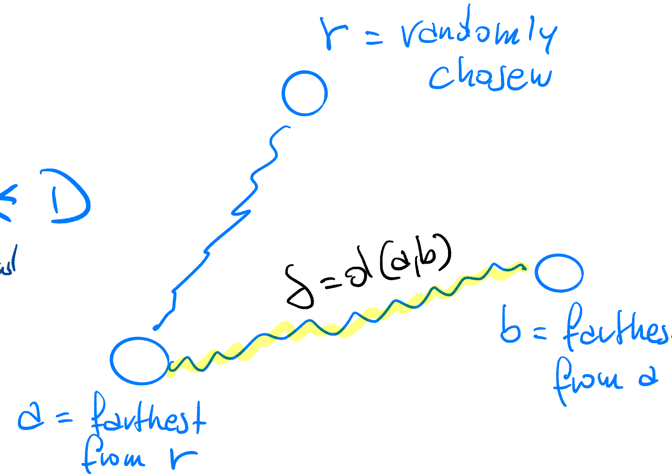
2-SWEEP

1. choose $r \in V$ uniformly at random
2. a = the last visited node in BFS(r) (a = farthest from r)
3. b = the last visited node in BFS(a) (b = farthest from a)
4. $\delta = d(a,b)$

We update L and U as follows:

- $L = \max\{L, d(r,a), \underset{\delta}{\underbrace{d(a,b)}}\} \leq D$

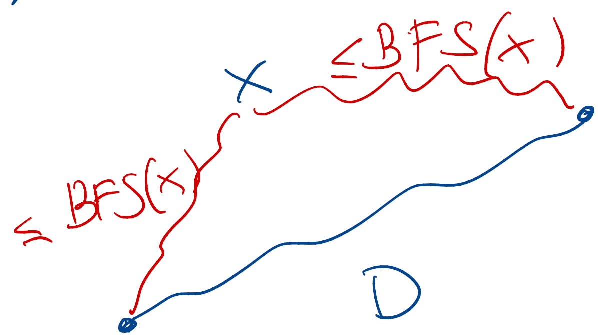
  obs. any BFS( ) is a lower bound for D



- $U = \min\{U, 2d(r,a), 2d(a,b)\} \geq D$

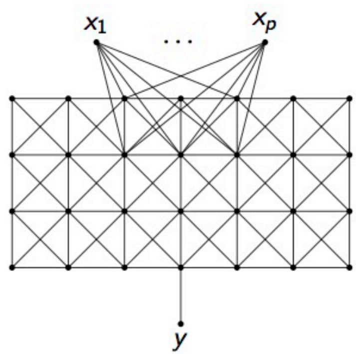
  obs triangle inequality

  $D \leq 2\ \mathrm{BFS}(x)$



No guarantee that $L = U$ (even repeating few more times)

Bad cases for 2-Sweep



In this graph, the heuristics gives a diameter far from D

In this modified grid with k rows and $1 + 3k/2$ columns. The algorithm returns $k + 1$. The diameter of the network is instead $3k/2$.

However, 2-SWEEP works amazingly well in real-world networks

Network name	D	Numb. of runs (out of 10) in which $LB = D$	Worst LB found
Wiki-Vote	9	10	9
p2p-Gnutella08	19	9	**18**
p2p-Gnutella09	19	9	**18**
p2p-Gnutella06	19	10	19
p2p-Gnutella05	22	9	**21**
p2p-Gnutella04	25	7	**22**
p2p-Gnutella25	21	8	**20**
p2p-Gnutella24	28	10	28
p2p-Gnutella30	23	2	**22**
p2p-Gnutella31	30	9	**29**
s.s.Slashdot081106	15	10	15
s.s.Slashdot090216	15	10	15
s.s.Slashdot090221	15	10	15
soc-Epinions1	16	9	**15**
Email-EuAll	10	10	10
soc-sign-epinions	16	10	16
web-NotreDame	93	10	93
Slashdot0811	12	10	12
Slashdot0902	13	3	**12**
WikiTalk	10	9	9
web-Stanford	210	10	210
web-BerkStan	679	10	679
web-Google	51	10	51

Whenever $L \neq D$, it is very close (typically $L = D - 1$)

Network name	D	Numb. of runs (out of 10) in which $LB = D$	Worst LB found
wordassociation-2011	10	9	**9**
enron	10	10	10
uk-2007-05@100000	7	10	7
cnr-2000	81	10	81
uk-2007-05@1000000	40	10	40
in-2004	56	10	56
amazon-2008	47	10	47
eu-2005	82	10	82
indochina-2004	235	10	235
uk-2002	218	10	218
arabic-2005	133	10	133
uk-2005	166	10	166
it-2004	873	10	873

We saw in the last class that deciding $D=2$ or $D=3$ in $\tilde{O}(n^{2-\varepsilon})$ time cannot be done unless SETH is false

What if we want to find an approximated $\tilde{D}$ with a guarantee?

<u>Ideal situation</u>: for any $0<\beta<1$, we would like to have

$$(1-\beta)D \leq \tilde{D} \leq D$$

This is not possible for any such β with $\boxed{(1-\beta) > \frac{2}{3}}$

Indeed this allows us to decide whether $D=2$ or $D=3$.

- $D=2 \Rightarrow \tilde{D} \leq D = 2$
- $D=3 \Rightarrow 2 = \frac{2}{3}D < (1-\beta)D \leq \tilde{D}$

Hence, given G, if we run the approximation $\tilde{D}$ for such β we say that $D=2$ if $\tilde{D} \leq 2$; we say that $D=3$ if $\tilde{D} > 2$

Hence, the best option is to get $(1-\beta) = \frac{2}{3}$

We see a randomized approximation algorithm for the diameter D that finds $\tilde{D}$ such that $\boxed{\frac{2}{3}D \leq \tilde{D} \leq D}$ w.h.p. in $O(n\sqrt{n}\lg n)$ time

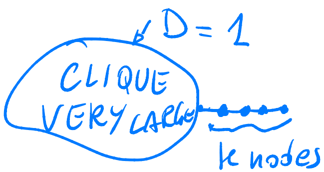
$G=(V,E)$, $n=|V|$, $m=|E|$

IDEA: Find a "small" subset $Z \subseteq V$, where $|Z| = O(\sqrt{n}\lg n)$,

and $\tilde{D} = \max_{u \in Z} BFS(u)$, so that $\underbrace{\frac{2}{3}D \leq \tilde{D}}_{\text{non-trivial part}} \leq D$.

$$D = \max_{\substack{u,v \in V \\ u \neq v}} d(u,v)$$

this a worst-case measure, different from the average distance: if we want a randomized algorithm for the diameter, sampling should be more sophisticated.



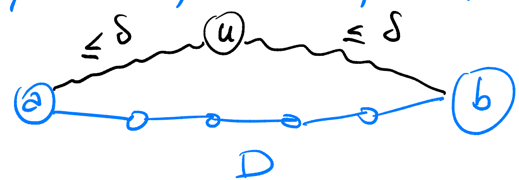
$\Rightarrow D = 1 + k$ but $|V| \gg k$

average distance ~ 1

3) Q.: Can we have a sort of PTAS for D?
That is, can we approximate D as much as we want within a given probability? NO

obs If we run BFS(u) from an node u, letting $\delta = ecc(u) = \max_{v \in V, v \neq u} d(u,v)$, then $D \leq 2\delta$

Two diametral nodes, a and b, that is, $d(a,b) = D$



$\Rightarrow$ TRIANGLE INEQUALITY $D \leq 2\delta$

The argument is the following:

Suppose by contradiction that we can get an approximation $\tilde{D}$ s.t. $(1-\varepsilon)\tilde{D} \leq D$ for any $\varepsilon > 0$

Recall that deciding whether the diameter $D=2$ or $D=3$ cannot be done in $O(n^{2-\varepsilon})$ unless SETH is false

Let's see what happens to $\tilde{D}$

unless SETH is false

- $D=2 \Rightarrow \tilde{D} \leq 2$
- $D=3 \Rightarrow \tilde{D} \geq (1-\alpha)D$

This leads to a contradiction when $(1-\alpha)D > 2$

Indeed let's compute $\tilde{D}$ in $O(n^{2-\varepsilon})$ time (for some ε)

- if $\tilde{D} \leq 2$ we claim that $D=2$
- if $\tilde{D} > 2$ we claim that $D=3$

Hence, we cannot take α as small as possible

$\Rightarrow$ since $D=3$ creates the issue, it should be $(1-\alpha)D \leq 2$

$\underset{D=3}{\Rightarrow} \Rightarrow (1-\alpha) \leq \frac{2}{3} \Rightarrow$ the best we can hope for is $\tilde{D} \geq \frac{2}{3} D$

We thus discuss a randomized algorithm that finds an approximation $\tilde{D}$ s.t.

$$\frac{2}{3}D \leq \tilde{D} \leq D$$

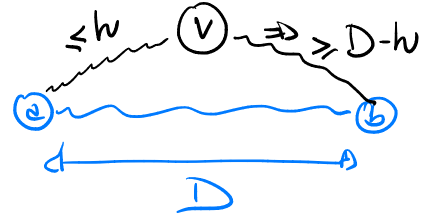
in $O(\underbrace{n\sqrt{n}\log n}_{\text{sample size}})$ time w.h.p.

G is sparse
$m = |E| = O(n)$

MAIN IDEA: Find a suitable sample of $O(\sqrt{n}\log n)$ nodes, such that one of them has eccentricity $\geq \frac{2}{3}D$

obs 1 We implicitly use the triangle inequality: $d(x,y) \le d(x,z) + d(z,y) \quad \forall z$

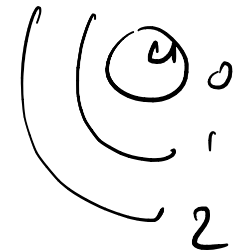
diametral nodes $a, b \in V$



otherwise $d(a,b) < D$

obs 2 Truncated BFS(u)

$N_k(u)$ = the first k nodes discovered by BFS(u)



$N_k(u)$ can be computed in $O(k^2)$ time

Indeed, given any node visited during BFS, if its adjacency list is $\le k+1$, we can traverse all of it; otherwise, even though the adjacency list can be longer than $k+1$, we surely stop at its first $k+1$ elements (we have to find just the first k in BFS).

$k = \sqrt{n}$

sparse $G=(V,E)$

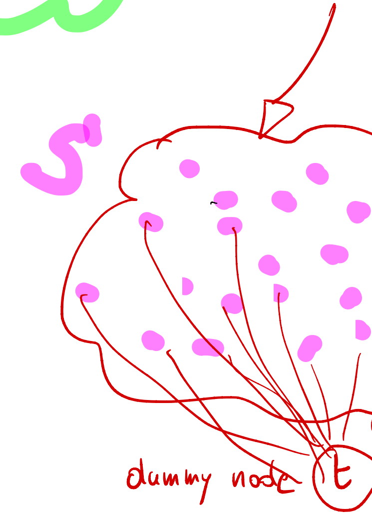
Here is the approximation algorithm:

1. S = set of $\alpha \frac{n}{k} \ln n$ ($\alpha > 2$) nodes randomly chosen from V in a uniform way

2. w = the last visited node from $BFS(S)$ (i.e. the farthest node from S)

3. $Z = S \cup N_k(w)$

<u>return</u> $\tilde{D} = \max_{u \in Z} ecc(u)$ [i.e. $BFS(u)$]



$BFS(t)$

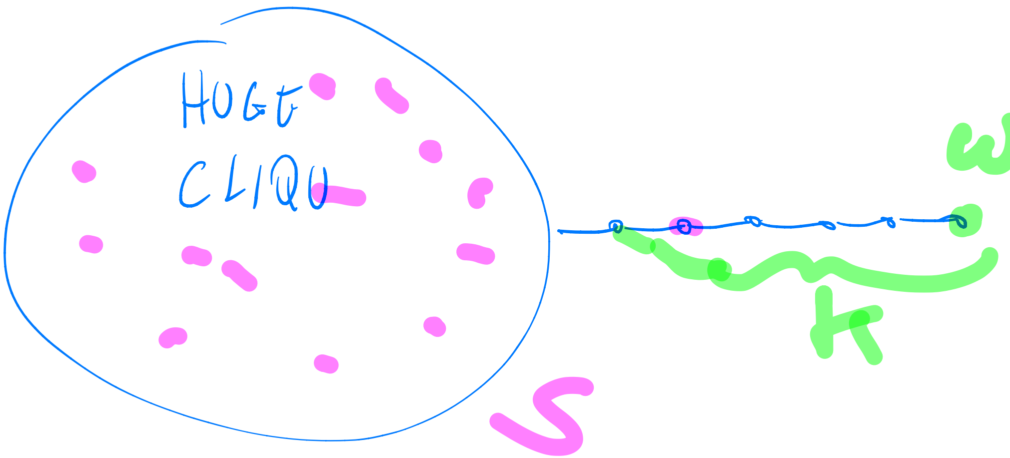
$G' = (V \cup \{t\}, E \cup \{(t,s) : s \in S\})$

TOTAL TIME = cost(1) + cost(2) + cost(3)

$$= O(|S|) + O(n) + O(|Z| \cdot n)$$

$$= O\left(\alpha \frac{n}{k} \ln n + n + \left(\alpha \frac{n}{k} \ln n + k\right) \cdot n\right) \overset{k=\sqrt{n}}{=} O(n\sqrt{n} \log n) \text{ time}$$

Intuition:



Why does it work?

a) S is a $\boxed{\text{hitting set}}$ for any $N_k(u)$, $u \in V$ w.h.p.

(we will use this when $u = w$)

↳ that is, $S \cap N_k(u) \neq \emptyset$

b) Let $D = 3h + z$, where $z \in \{0, 1, 2\}$

$$\left.\begin{array}{ll} 2h + z \leq \tilde{D} \leq D & \text{w.h.p. when } z = 0,1 \\ 2h + 1 \leq \tilde{D} \leq D & \text{w.h.p. when } z = 2 \end{array}\right\} \Rightarrow \frac{2}{3} D \leq \tilde{D}$$

Let's see α!

$$\Pr[S \text{ is a hitting set for } N_k(u), \text{ any } u \in V] =$$

$$1 - \underbrace{\Pr[\exists u \in V : S \cap N_k(u) = \emptyset]}_{} =$$

$$p = \Pr[\underbrace{S}_{\text{random choices here}} \cap \underbrace{N_k(u)}_{\text{target of k nodes}} = \emptyset] = \Big(\underbrace{\frac{n-k}{n}}_{\text{prob. that a node in S is \underline{not} in the target of k nodes}}\Big)^{\overset{\text{\# experiments}}{\downarrow}\,|S|} = \left(1 - \frac{k}{n}\right)^{|S|}$$

$$= \left(e^{-\frac{k}{n}}\right)^{|S|} = e^{-\frac{\cancel{k}}{\cancel{n}} \cdot \alpha \frac{\cancel{n}}{\cancel{k}} \ln n} = \frac{1}{n^{\alpha}}$$

$(1-x) \sim e^{-x}$

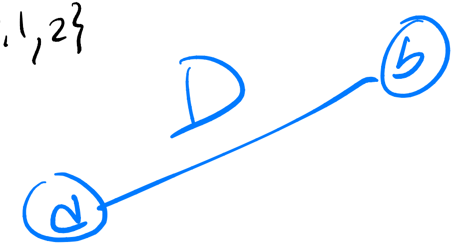
$x = \frac{k}{n}$

$|S| = \alpha \frac{n}{k} \ln n$

$$= 1 - np = 1 - \frac{1}{n^{\alpha-1}} \quad \text{w.h.p.} \quad \alpha - 1 > 1 \Rightarrow \alpha > 2$$

UNION BOUND ON $u \in V$

Let's see $\underline{b)}$ Recall $D = 3h+z$, $z \in \{0,1,2\}$



a,b = diametral nodes, that is, $d(a,b) = D$

w = found in step $\boxed{2}$

We have two cases

CASE b.1 : $d(w,S) \leq h$ (hence, every node is at distance $\leq h$ from S)

Hence $d(a,S) \leq h$ as w is the farthest from S

Now consider $x \in S$ that is closest to a

$\Rightarrow BFS(x) \geq d(x,b) \geq D-h$ by triangle inequality as $d(a,x) \leq h$

$\Rightarrow BFS(x) \geq D-h = 2h+z$

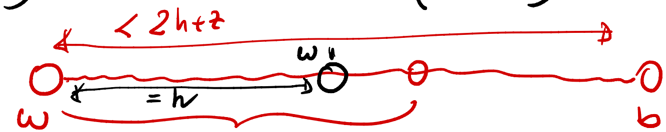
$D = 3h+z$

The rationale here is telling when a random sample S is enough to approximate

CASE b.2 : $d(w, S) > h$ (i.e. every node in S is at distance $> h$ from w)

- $BFS(w) \geq 2h + z$ we are done ?

- $BFS(w) < 2h + z \Rightarrow d(w, b) < 2h + z$



first k elements
(intersection of this path with $N_k(w)$)

we want to show that $\exists\, w'$ along this path s.t. $d(w, w') = h$

Since S is a hitting set w.h.p., $S \cap N_k(w) \neq \emptyset$

$\Rightarrow \exists\, x \in S \cap N_k(w)$ s.t. $d(w, x) > h$ as

$\Rightarrow N_k(w)$ contains all nodes at distance h

(so $w' \in N_k(w)$ and $d(w, w') = h$)

w.h.p. $\exists\, w' \in N_k(w)$ along the path from w to b s.t. $d(w, w') = h$

$[\text{if } d(w,b) < h \Rightarrow w' = b]$

$$\overset{TI}{\Longrightarrow} d(w', b) < h + z \overset{TI}{\Longrightarrow} d(w', a) \geq D - d(w', b) \geq D - (h + z - 1) = 2h + 1$$

$D = 3h + z$

The rationale here is telling why BFS from a node in $N_k(w)$ makes sense.

Hence, the choice of z $\boxtimes$